

Functional Prototype

Design Documentation

- 1) Original design
 - a) Tensioned fishing line supported by springs, bearings, pulleys.
 - b) Bearings and springs mounted on 3/8in shafts with shaft collars.
 - c) Mounted on a frame consisting of 2 custom-fabricated plates held apart by 3/8in standoffs
 - d) The entire assembly uses 4-40 screws.
- 2) Issues and improvements (Chronological order)
 - a) 3d printed-pulleys were not delivered on time
 - b) Shafts were slightly larger than 3/8in nominal diameter - Shafts had to be sanded down.
 - c) The frame was poorly cut and screw holes were not in line - Use laser cut aluminum instead of a wooden frame.
 - d) Tensioned fishing line kept slipping off of bearings - Use 3d-printed flanged pulleys
 - e) Linear coil springs tangled easily and had to be manually adjusted - Create 3d-printed channel to serve as guide rail for springs.

Parts:

Bearing: MMC 6383k32 x2 (\$26.14)

3/8in diam shaft: MMC 8974T24 2ft x1 (\$4.49)

Shaft Collars: MMC 9414T8 x 8 (\$16.80)

Springs: MMC 9293K46 x 2 mounted on 8mm shaft (\$26.92), Retracting reels like tape measure

Two custom-fabricated plates (wood/aluminum) separated by 3/8ths standoffs w/ 4-40 screws

Standoffs: MMC 91115A509 X12 (\$15.36)

4-40 [Screw](#):

<https://www.mcmaster.com/92196A108/> (\$6.42)

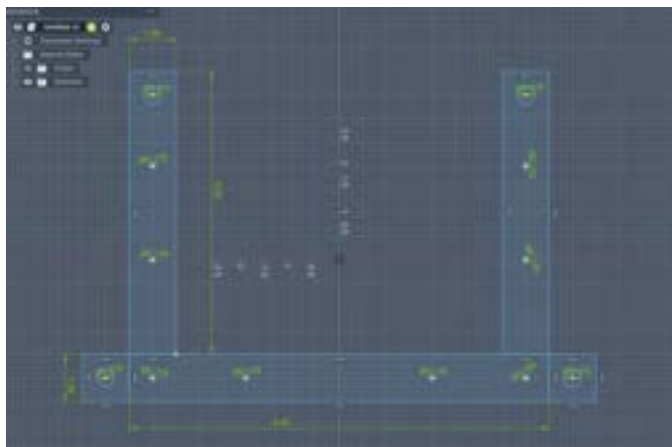
Multipurpose aluminum (3ft):

<https://www.mcmaster.com/8975K54-8975K881/> (\$14.84)

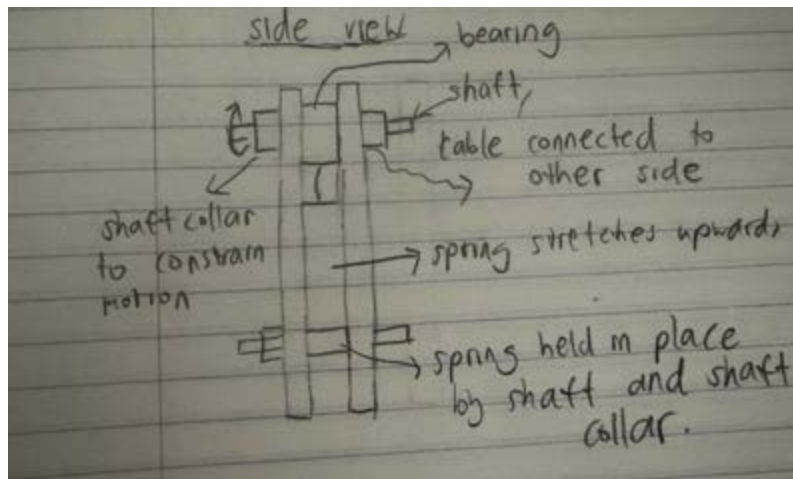
[Locking pins for adjustable hinge \(2\)](#): (\$12)

<https://www.mcmaster.com/90293A313/> (\$64.52)

Illustration:



Design Intent & Functionality:



Assembly Process:

- Sketch outline of each side of scraper assembly on wood plates
- Sketch drill holes for screws and shafts on wood plate assemblies
- Drill press holes for screws and shaft on wood plate assemblies
- Cut out the wood plate assembly with bandsaw
- Assemble scraper mechanism using shafts and screws
 - Springs and rollers are on the shafts
 - Screws go through smaller screw holes
- Attach scraper mechanism to extendable rod

Design Tests



Chris sketching the design on wood plates from digital sketch



Fatima sketching the design on wood plates from digital sketch



Fatima using a mill to cut holes into the wood plates



Fatima cutting the wood plates



Dylan attaching bearings onto the wood plates



Dylan inserting shafts into the wood plate



Half of the wood plates assembly



Full wood plates assembly completed



Scraper head attached on the pole

Design Tests

Motion and Interference

- String should be able to “bend” to conform to area with no problem while maintaining force
- The mechanism successfully achieved a motion range of approximately **0° to 110°**, slightly below the target of 120°

- The string was able to bend around contours down to an estimated **radius of ~2–3 cm** without significant loss of tension
- Minor friction was observed at extreme angles, but no jamming occurred during the first **8 cycles**
- After repeated motion, slight resistance increased by about **10–15%**
- **Implication: Hard-install a maximum range of motion to minimize damage by extreme bending, and improve the consistency of motion by making better spring supports and using a roller with a built in track for the cable.**

Repeated Use

- The system was tested for 10 cycles total
- Performance remained consistent for the first 5 cycles
- After cycle 5, degradation began as the cable started to slip off the bearings and lose consistent tension, leading to less resistance against the motion.
- **Implication: The cable needs to be attached to our bearings more securely**

Assembly and Fastening:

- Simulate pushing / pulling motion
- Assembly withstood forces of approx 5-8N
- **Implication: Either need instructions to prevent excessive force usage or increase the amount of force able to be withstood (higher quality parts, more accurate assembly)**

Stability and Support:

- When extending pole to different lengths, the user shouldn't have issue simulating the same range of motion
- Noticed that at long lengths the pole becomes unstable and harder to control, there is instability caused by the increased torque created when force is applied at a greater distance from the base
- **Implication: Need counterweight to prevent tipping of pole at applied force**

Success Criteria

Context:

This prototype is intended to act as a flexible scraper mechanism that can conform to curved surfaces while maintaining consistent cable tension and reliable motion. Success will be measured by how well the mechanism performs repeated scraping motion, maintains alignment of its components, and remains stable and usable when mounted on an extendable pole.

Criteria:

1. Range of Motion

The scraper mechanism should achieve a controlled cable stretch range of 3 inches from initial height without jamming, cable derailment, or permanent deformation of any component.

- Reaching **3 inches** is a reasonable goal and will allow the cable to conform around the outer shape of the egg masses
- Any stretch length greater than 3 inches is not a high priority because we also do not want our device to stretch too much since having high tension was also an essential aspect of our design

- This will be measured by manually recording the maximum length reached before interference or excessive resistance occurs.

2. Cable Retention and Tracking

During operation, the cable must remain fully seated on its pulley/bearing path for at least **20 consecutive motion cycles** with zero derailments.

- This criterion directly addresses the already observed issue of the fishing line potentially slipping off the bearings but will also ensure our overall structure and assembly can hold itself together under repeated loadings
- Achieving a zero derailment rate over more than 20 tested cycles is a high priority because maintaining proper cable tracking is critical to both the functionality and durability of the mechanism.

3. Pole-Mounted Usability and Stability

When attached to the extendable rod at its intended operating length, the system should remain controllable enough that **5 different users** can complete a full scraping motion without excessive wobble, tipping, or loss of alignment.

- Quantitatively, the pole should remain stable enough to allow **10 consecutive successful scraping strokes** at full extension without much struggle .
- This criteria will be quantitatively evaluated by asking different students in our lab section and people visiting us during showcase day
- Improved stability at extremely long extension is a middle to high priority because while we believe it is crucial to make our device user friendly, we don't expect the spotted lantern fly egg masses to be so high that stability will be an extreme issue.

The above listed criterias have the potential to be demonstrated during the exhibit day showcase through a live, interactive test in which users can operate the device to perform repeated scraping motions on mock egg masses. We have chosen to focus on criteria #2 as mechanical reliability is our primary priority which would allow the device to function smoothly under repeated use. Although it would be impossible to simulate what egg masses are like in real life, having a mock egg mass that one group member will hold up while a test user lightly “pushes” to “scrap” away will allow them to experience tension forces at work and see first hand how reliable our device is.